

Fundamentals of Cyber Security and Digital Forensics

Module: 2

Priyanka Shingane

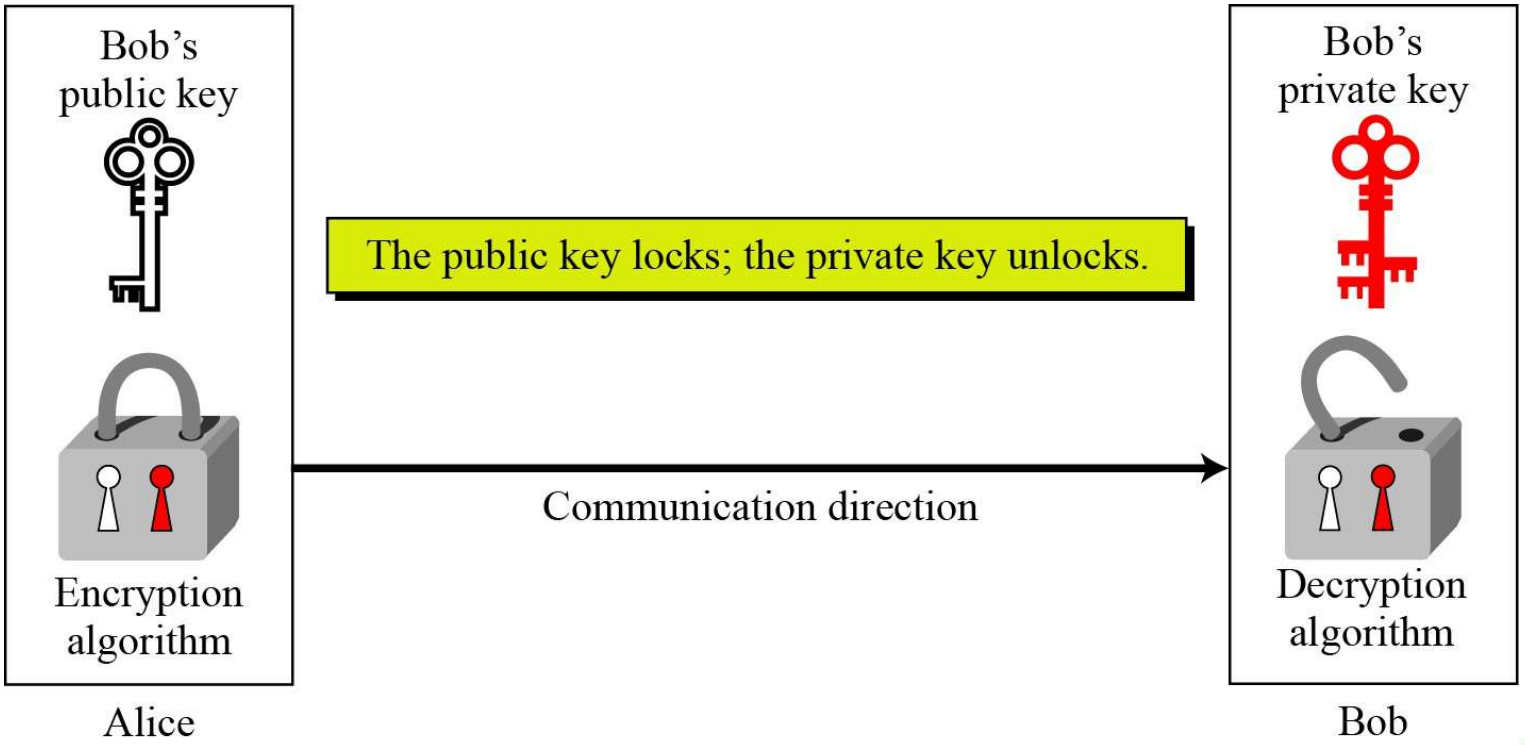
Module 2:

Lecture 11: Asymmetric Key Cryptography-RSA



Principles of Public Key Cryptography

It is a process that uses a pair of related keys -- one public key and one private key -- to encrypt and decrypt a message and protect it from unauthorized access or use



Asymmetric Key Cryptography:

- Asymmetric cryptography, also known as public-key cryptography, operates using two distinct keys: a public key and a private key.
- The public key is openly shared with everyone, while the private key remains confidential.
- This approach enables secure communication without the need to exchange secret keys directly.



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Principles of Public Key Cryptography

Plaintext/Ciphertext

Unlike in symmetric-key cryptography, plaintext and ciphertext are treated as integers in asymmetric-key cryptography.

$$C = f(K_{public}, P) \qquad P = g(K_{private}, C)$$



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RSA Algorithm Overview:

- RSA (Rivest–Shamir–Adleman) is a widely used asymmetric encryption algorithm.
- It relies on the mathematical difficulty of factorizing large integers.
- The security of RSA hinges on the fact that it is computationally infeasible to reverse the process of multiplying two large prime numbers.

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Key Generation

- **Public Key (n, e):**
  - Select two large prime numbers, say **P** and **Q**.
  - Compute **n = P × Q**.
  - Choose a small integer **e** such that **1 < e < Φ(n)** (where **Φ(n)** is Euler's totient function).
  - The public key consists of **n** and **e**.
- **Private Key (d):**
  - Calculate **Φ(n) = (P - 1) × (Q - 1)**.
  - Find an integer **k** such that **d = (k × Φ(n) + 1) / e**.
  - The private key is **d**.

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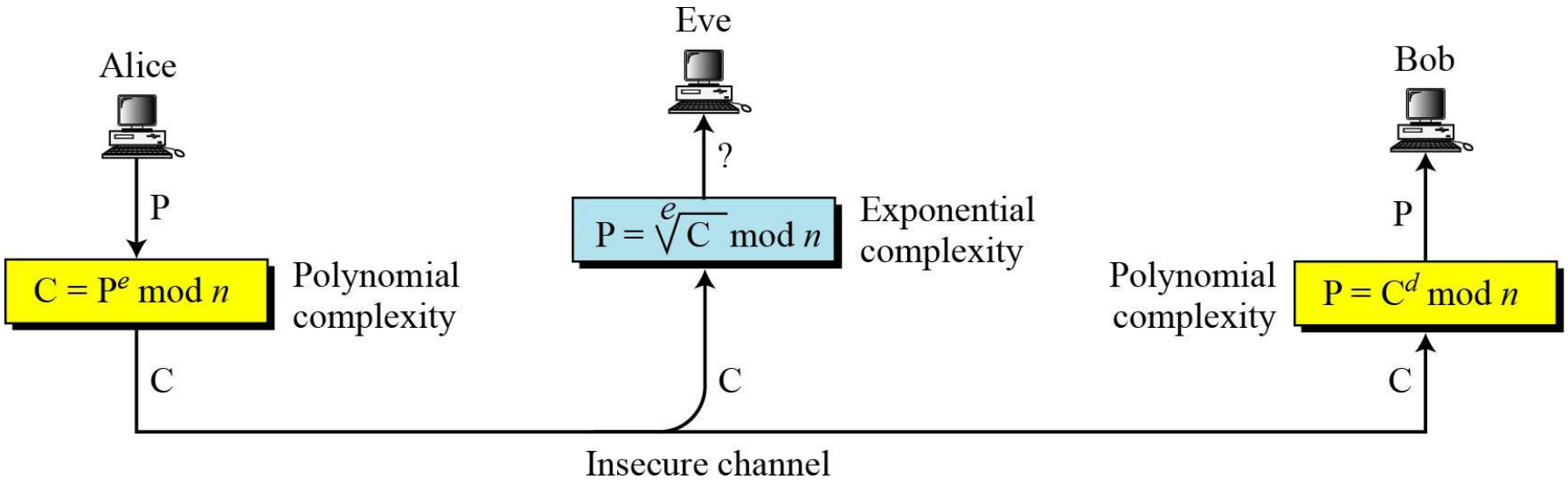
Encryption and Decryption in RSA

Encryption and Decryption:

1. **Encryption (Sender's Side):**
    1. Convert the plaintext message (e.g., "HI") into numerical values (e.g., H = 8, I = 9).
    2. Compute the ciphertext **c = (message^e) mod n**.
1. **Decryption (Receiver's Side):**
    1. To decrypt, compute **plaintext = (c^d) mod n**.
    2. The original message is recovered.

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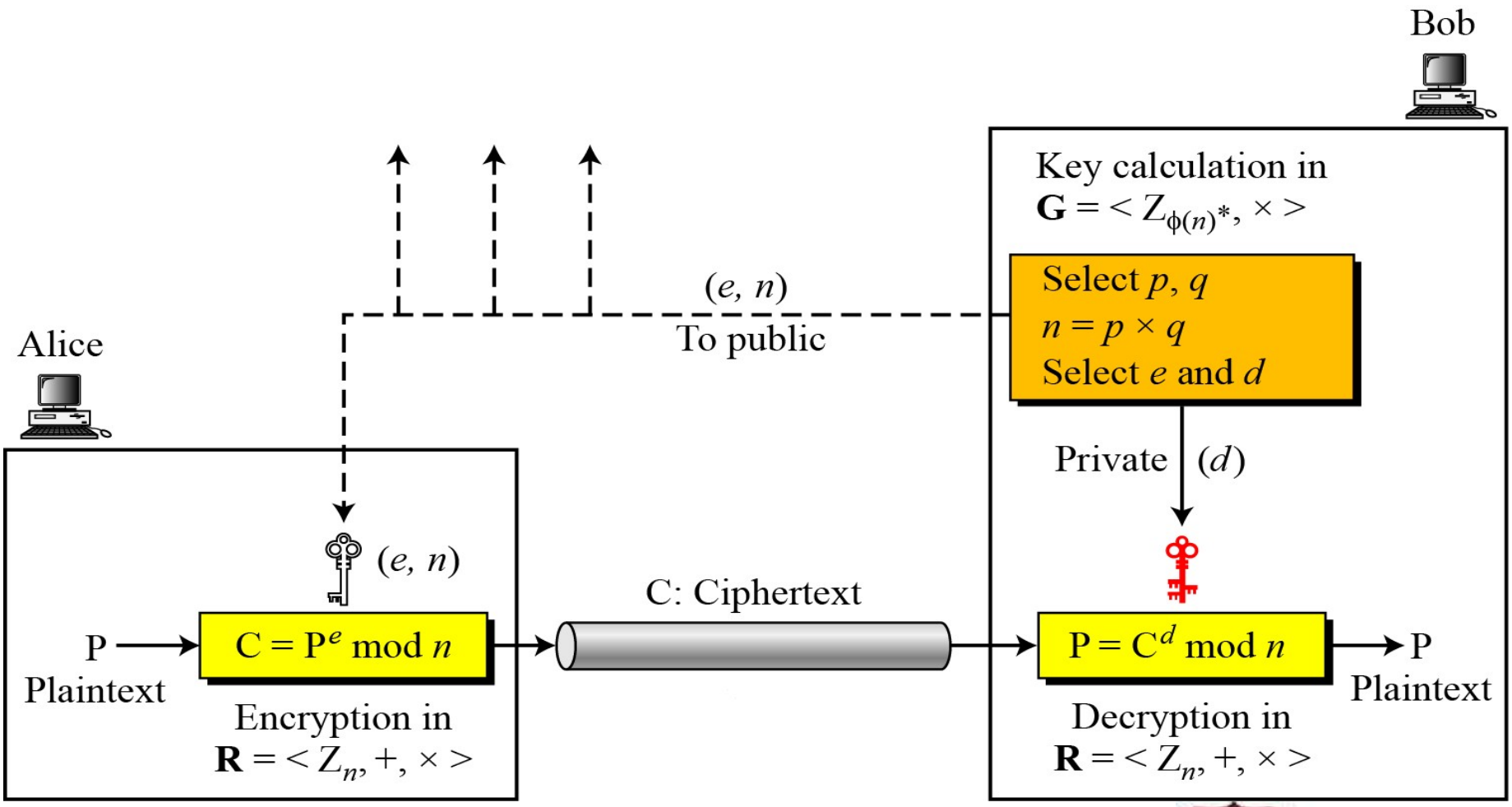
RSA CRYPTOSYSTEM



RSA uses modular exponentiation for encryption/decryption; To attack it, Eve needs to calculate $\sqrt[e]{C} \bmod n$.

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RSA CRYPTOSYSTEM



Strength and Key Sizes and Application in Cyber Defense

- RSA keys are typically 1024 or 2048 bits long.
- Experts recommend using longer key sizes (e.g.2048 bits) for enhanced security.
- While 1024-bit keys are still secure, there are concerns about their longevity.

Application in Cyber Defence:

- RSA plays a crucial role in:
  - **Secure communication**: Encrypting data during transmission.
  - **Digital signatures**: Authenticating messages and verifying sender identity.
  - **Key exchange protocols**: Establishing secure channels between parties.

The strength of RSA lies in its mathematical foundation, making it a vital tool in safeguarding sensitive information across the digital landscape.



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RSA CRYPTOSYSTEM

RSA_Key_Generation

{

Select two large primes p and q such that $p \neq q$.

$n \leftarrow p \times q$

$\phi(n) \leftarrow (p-1) \times (q-1)$

Select e such that $1 < e < \phi(n)$ and e is coprime to $\phi(n)$

$d \leftarrow e^{-1} \bmod \phi(n)$ // d is inverse of e modulo $\phi(n)$

Public_key $\leftarrow$ (e, n) // To be announced publicly

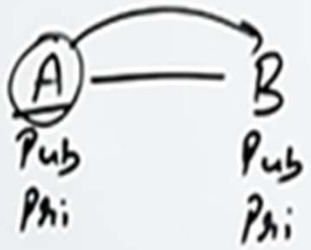
Private_key $\leftarrow d$ // To be kept secret

return Public_key and Private_key

}

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In a RSA cryptosystem a particular A uses two prime nos $p=13$ and $q=17$ to generate his public and private keys. If the Public key of A is 35. Then the Private Key of A is ____ (A) 11 (B) 13 (C) 16 (D) 17



1. Choose two different large random prime no.
2. Calculate $n = p * q$
3. Calculate $\phi(n) = (p-1) * (q-1)$
4. Choose 'e' such that $1 < e < \phi(n)$
   e is coprime to $\phi(n)$, $\gcd(e, \phi(n)) = 1$
5. Calculate d, such that $de \equiv 1 \bmod \phi(n)$
6. Public key 'e' Private key 'd'

$1 < e < 192$

$p = 13$, $q = 17$

$n = p \times q = 13 \times 17 = 221$

$\phi(n) = (p-1) \times (q-1) = 12 \times 16 = 192$

$e = 35$ $\gcd(35, 192) = 1$

$de = 1 + K\phi(n)$

$d = \frac{1 + K \cdot \phi(n)}{e}$ $K = 0, 1, 2 \ldots$

$= \frac{1 + 0 \times 192}{35} = \frac{1}{35} = 0.$

$\frac{1 + 1 \times 192}{35}$

$= \frac{193}{35} = 5.5$

$d = \frac{1 + 2 \times 192}{35} = \frac{385}{35} = 11$

$d = 11$

TAXONOMY OF POTENTIAL ATTACKS ON RSA

Plain text Attack: Plain text attacks are classified into three categories

- Short message attack: In this type of attack, the assumption is that the attacker knows some blocks of the plain text message. If an attacker knows some block of plain text, then he could try to encrypt the blocks of plain text using the information and try to convert it into cipher text. To prevent a short message attack, we can use the padding bits for encryption.
- Cycling attack: In cycling attack, the reverse process is done. An attacker assumes that the ciphertext is formed using some permutations operations. If the attacker assumption becomes true, then he can try the reverse process to obtain the plain text from the cipher text.
- Unconcealed message attack: In some rare cases, it is found that some encrypted cipher text is the same as the plain text i.e original text. This means that the plain text is not hidden. Such type of attack is called an unconcealed message attack

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TAXONOMY OF POTENTIAL ATTACKS ON RSA

Chosen cipher Attack: In this type of attack, the attacker can find out the plain text from cipher text using the extended Euclidean algorithm.

Factorization Attack: If attacker is able to find out P and Q from N=PXQ, then the attacker can find out the private key. Assuming that if N is at least 300 digits long then attacker cannot find P and Q.

Attacks on Encryption key: Small value of E can lead to attack on encryption key. Hence it is recommended to use E as $2^{16}+1=65537$ or a value closer to this.

Attacks on the Decryption Key:

- Revealed decryption exponent attack: If attacker somehow guess decryption key D, not only the cipher text generated by encryption the plain text with corresponding encryption key is in danger, but even future messages are also in danger. So, it is advised to take fresh values of two prime numbers (i.e; P and Q), N and E.
- Low decryption exponent attack: If we take smaller value of D in RSA this may occur so to avoid this take value of D = 2^16+1(atleast).

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Principles of Public Key Cryptography

Characteristic	Symmetric Cryptography	Asymmetric Cryptography
Key used for encryption/decryption	Same key is used	One key is used for encryption and another for decryption
Speed of encryption/decryption	Very fast	Slower
Size of resulting encrypted text	Usually same as or less than the original plaintext size	More than the original plaintext size
Known keys	Both parties should know the key in symmetric key encryption	One of the keys is known by the two parties in public key encryption
Usage	Confidentiality	Confidentiality, Digital signature



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Module 2:

Lecture 12: Diffie-Hellman Key Exchange Protocol



Step-by-Step Process:

The **Diffie-Hellman key exchange protocol** is essential for secure communication in scenarios where two parties need to establish a shared secret key over an insecure channel

- Alice and Bob agree on a large prime number and a primitive root modulo number.
- Each secretly selects a private key (random number).
- They compute their public keys based on the prime, primitive root, and private keys.
- Alice and Bob exchange their public keys.
- Using the exchanged public keys and their own private keys, they derive a shared secret key.

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Diffie-Hellman:

- **Diffie-Hellman**:
  - Employs **exponential methods** for key generation.
  - Uses the same key for both encryption and decryption.

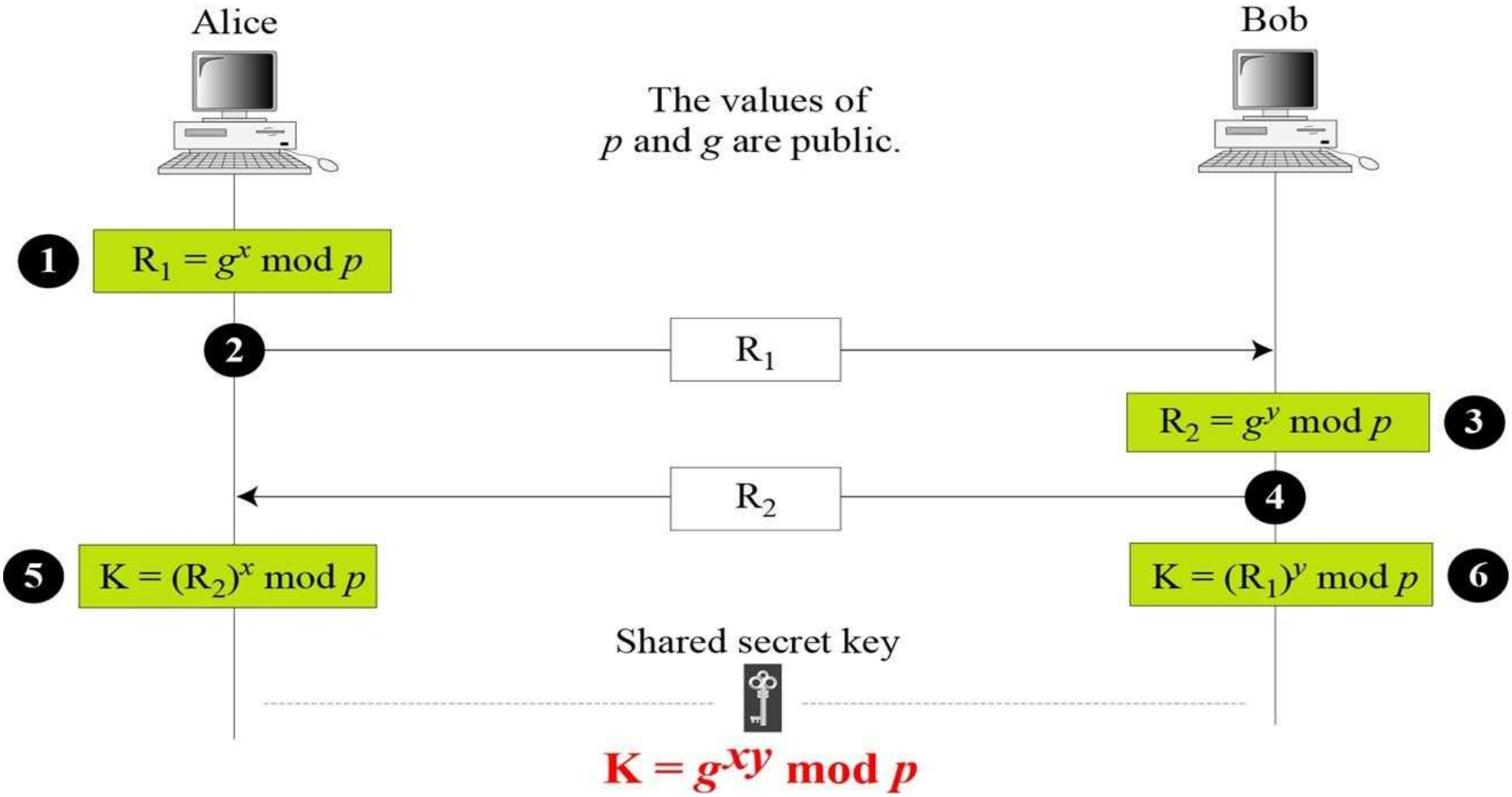
Application Focus:

- **Diffie-Hellman**:
  - Primarily used for **secure key exchange**.
  - Establishes a shared secret key between parties over an insecure channel.



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Diffie-Hellman Key Exchange Protocol



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Diffie-Hellman Key Exchange Protocol

Alice	**Bob**
Public Keys available = P, G	
Private Key Selected = a	Private Key Selected = b
Key generated = $R_1=G^a \bmod P$	Key generated = $R_2=G^b \bmod P$
Exchange of generated keys takes place	
Key received= R_2	Key received= R_1
Generated Secret Key = $K_a=(R_2)^a \bmod P = (G^b)^a \bmod P$	Generated Secret Key = $K_b=(R_1)^b \bmod P=(G^a)^b \bmod P$
Users now have a symmetric secret key to encrypt, $K_a=K_b=G^{ab} \bmod P$	

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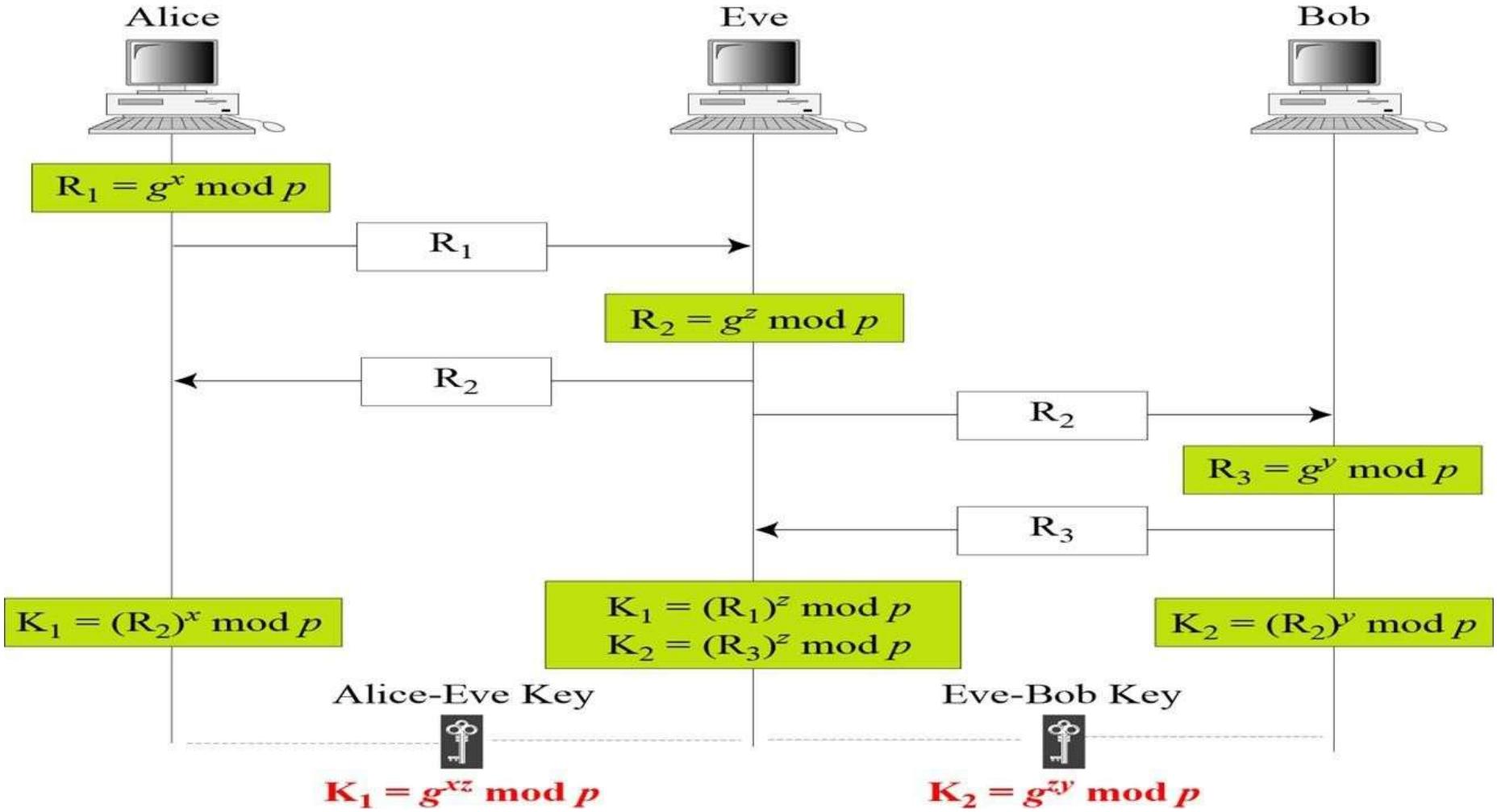
Diffie-Hellman Key Exchange Protocol

Let us give a trivial example to make the procedure clear. Our example uses small numbers, but note that in a real situation, the numbers are very large. Assume that $g = 7$ and $p = 23$. The steps are as follows:

1. Alice chooses $x = 3$ and calculates $R_1 = 7^3 \bmod 23 = 21$.
2. Bob chooses $y = 6$ and calculates $R_2 = 7^6 \bmod 23 = 4$.
3. Alice sends the number 21 to Bob.
4. Bob sends the number 4 to Alice.
5. Alice calculates the symmetric key $K = 43 \bmod 23 = 18$.
6. Bob calculates the symmetric key $K = 216 \bmod 23 = 18$.
7. The value of K is the same for both Alice and Bob; $g^{xy} \bmod p = 718 \bmod 35 = 18$.

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Man-in-middle Attack in DH



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Application in Cloud Security

Application in Cloud Security:

- **Secure Data Transmission**: In cloud environments, Diffie-Hellman ensures secure text transfer. Users can transmit secret messages within text files, encrypting them with a shared key. Even if the key is compromised, the Diffie-Hellman protocol renders it useless for hackers.
- **SSL/TLS**: Secure Sockets Layer (SSL) and Transport Layer Security (TLS) protocols use Diffie-Hellman to establish secure channels between clients and servers over the Internet.

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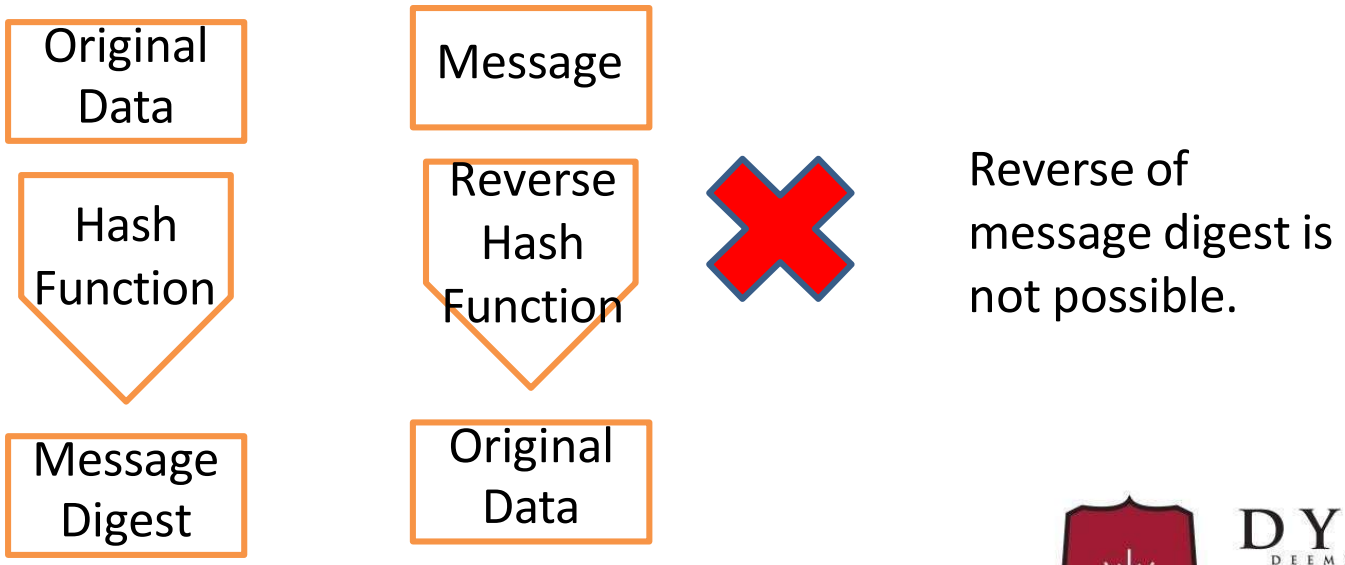
Module 2:

Lecture : Hashing Algorithm



Message Integrity

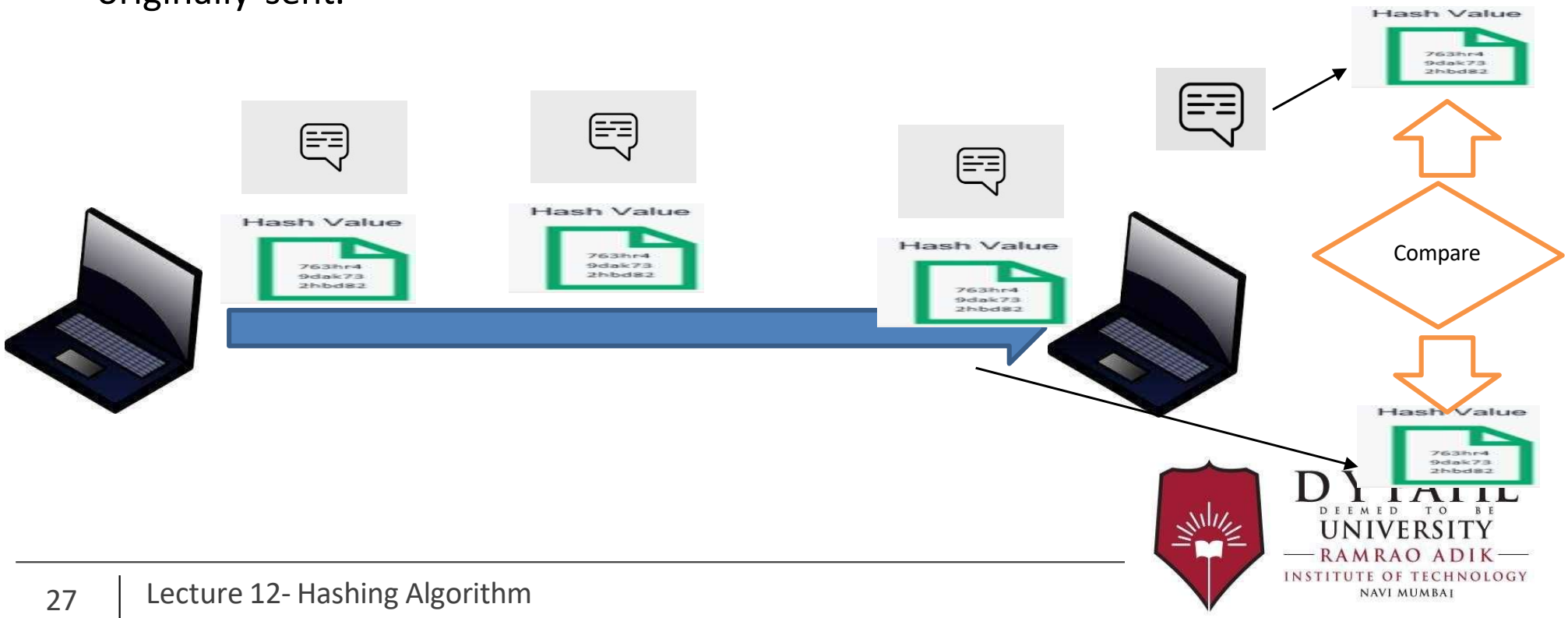
- Message integrity means that a message has not been tampered with or altered.
- The most common approach is to use a hash function that combines all the bytes in the message with a secret key and produces a message digest that is difficult to reverse.
- A message digest/hash is a fixed size numeric representation of the contents of a message, computed by a hash function.



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Message Integrity

- The sender will calculate a hash on the message, and include the digest with the message.
- On the other side, the receiver will independently calculate the hash on just the message, and compare the resulting digest with the digest which was sent with the message.
- If they are the same, then the message must have been the same as when it was originally sent.



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Applications of hashing :

- **Digital Signatures**: Hashing is used to generate digital signatures, ensuring data integrity and authenticity.
- **Message Authentication Codes (MACs)**:Hash functions create MACs for secure communication.
- **Key Derivation Functions**: Hashing helps derive cryptographic keys.
- **Blockchain Technology**: Hashing secures the integrity and consensus of blockchain data, including the **proof-of-work algorithm**.
- **Fraud Detection and Cybersecurity**: Hashing aids in detecting and preventing malicious activities, including intrusion detection and antivirus software.
- **Database Indexing**: Hashing is used to efficiently index and retrieve data in databases and other data storage systems. It allows for fast lookups and queries.
- **Password Storage**: Hashing is employed to securely store passwords. Instead of storing plain text passwords, systems store the hashed result of the password using a hash function. This enhances security and prevents exposure of sensitive information.

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MD5

- Designed in 1991 by ronald rivest
- MD5 is a cryptographic hash function algorithm that takes the message as input of any length and changes it into a fixed-length message of 16 bytes(128 bits).

- The output of MD5 (Digest size) is always 128 bits.

input Message



Output 128 bit of Digest

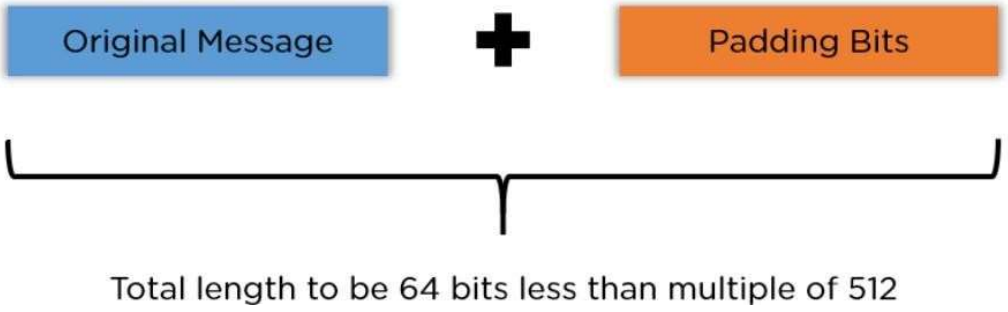
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MD5

MD5 algorithm follows the following steps

- Append Padding Bits: In the first step, we add padding bits in the original message in such a way that the total length of the message is 64 bits less than the exact multiple of 512. Suppose we are given a message of 1000 bits. Now we have to add padding bits to the original message. Here we will add 472 padding bits to the original message. After adding the padding bits the size of the original message/output of the first step will be 1472 i.e. 64 bits less than an exact multiple of 512 (i.e. 512*3 = 1536).

  Length(original message + padding bits) = 512 * i – 64 where i = 1,2,3 . . .



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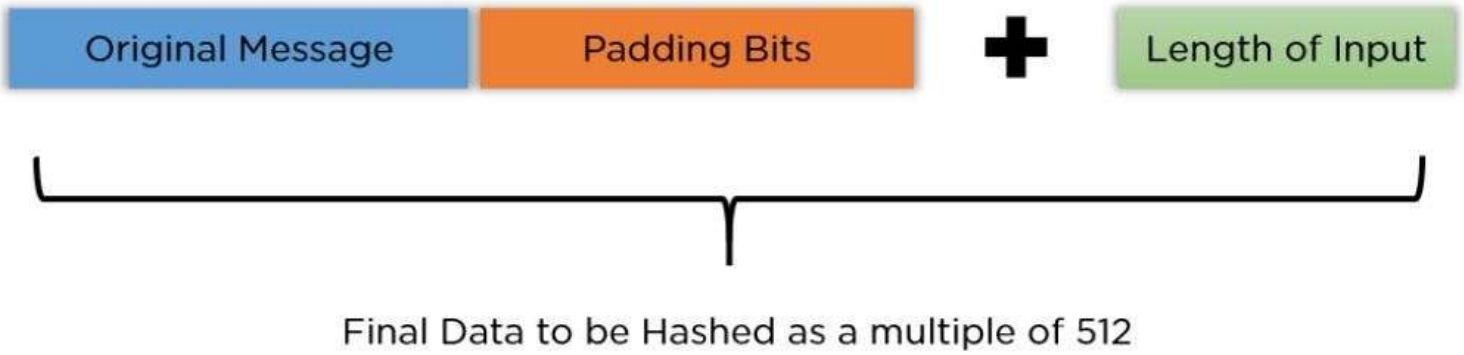
MD5

- Append Length Bits: In this step, we add the length bit in the output of the first step in such a way that the total number of the bits is the perfect multiple of 512. Simply, here we add the 64-bit as a length bit in the output of the first step.

  i.e. output of first step = 512 * n – 64

  length bits = 64.

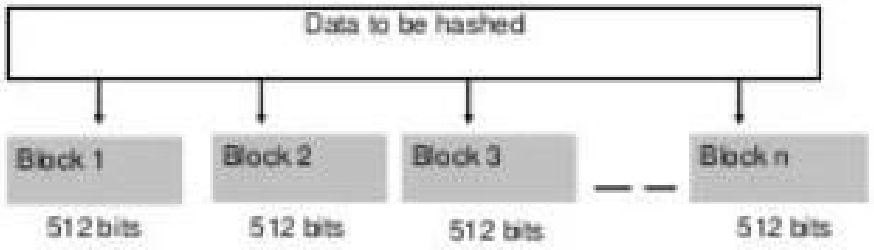
  After adding both we will get 512 * n i.e. the exact multiple of 512.



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MD5

- Divide the input into 512 bit blocks



- Initialize MD Buffer: The entire string is converted into multiple blocks of 512 bits each. You also need to initialize four different buffers, namely A, B, C, and D. These buffers are 32 bits each and are initialized as follows:

A = 01 23 45 67

B = 89 ab cd ef

C = fe dc ba 98

D = 76 54 32 10

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MD5

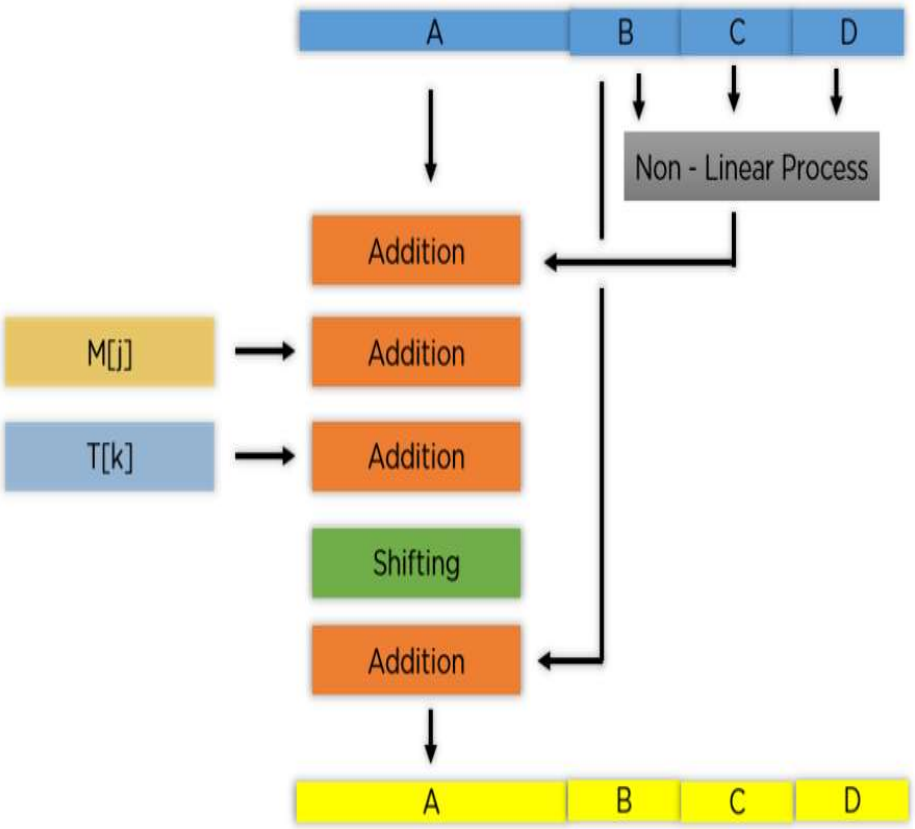
Process Each Block

- Each 512-bit block gets broken down further into 16 sub-blocks of 32 bits each.
- There are four rounds of operations, with each round utilizing all the sub-blocks, the buffers, and a constant array value.
- This constant array can be denoted as T[1] -> T[64].
- Each of the sub-blocks are denoted as M[0] -> M[15].

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MD5

- It passes B, C, and D onto a non-linear process.
- The result is added with the value present at A.
- It adds the sub-block value M[j] to the result above.
- Then, it adds the constant value T[k] for that particular iteration.
- There is a circular shift applied to the string.
- As a final step, it adds the value of B to the string and is stored in buffer A.



Mathematical expression of MD5

a=b+((a+ Process P(B,C,D) + M[i] + T[k])<<<s)

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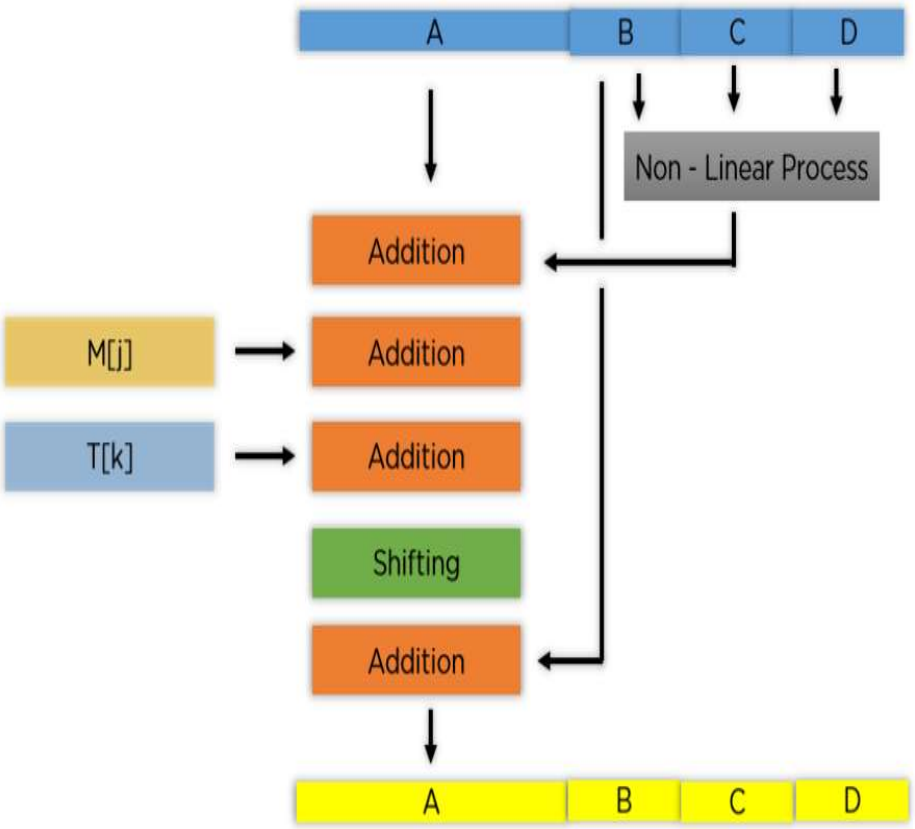
MD5

Round 1: (B AND C) OR ((NOT B) AND (D))

Round 2: (B AND D) OR (C AND (NOT D))

Round 3: B XOR C XOR D

Round 4: C XOR (B OR (NOT D))



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Advantages of MD5:

Simplicity and Speed:

- **Ease of Understanding**: MD5 is straightforward to comprehend, making it accessible to developers.
- **Fast Computation**: It efficiently generates hash values, making it suitable for quick operations.

Fixed-Length Hashes:

- MD5 produces a **fixed-length 128-bit hash** (16 bytes) regardless of the input size.
- This consistency simplifies comparisons and storage

Low Memory Requirements:

- **Efficient Integration**: MD5 requires relatively low memory for implementation.
- This makes it suitable for various applications without excessive resource usage

File Integrity Verification:

- **Checksums**: MD5 helps verify file integrity after transmission or download.
- By comparing the MD5 hash before and after transfer, users can detect any corruption or tampering

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Applications of MD5:

- **Data Verification**: MD5 hashes are employed to verify data integrity during transmission. By comparing the hash of the original data with the hash of the received data, we can detect any tampering or corruption.
- **Password Storage**: Historically, MD5 was used to store and verify user passwords. Now bcrypt and Argon2 are recommended.
- **Digital Signatures**: MD5 generates digital signatures, ensuring data authenticity and integrity.

File Integrity Checking:

- **After File Transfer**: The most common application of MD5 today is checking file integrity after a transfer. By generating an MD5 hash before and after transferring a file, any corruption or modification can be identified.
- **Ensuring File Consistency**: MD5 helps ensure that files remain consistent and unaltered.

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Applications of MD5:

Web Security:

- **Secure Passwords**: In web applications, MD5 was used to secure user passwords. However, due to its vulnerabilities, better alternatives (such as **bcrypt** or **Argon2**) are now recommended.
- **SSL Certificates**: MD5 was previously used in digital certificates, but its weaknesses led to its abandonment.

Network Security:

- **Intrusion Detection**: MD5 can be part of intrusion detection systems, verifying the integrity of system files.
- **Firewalls and Network Appliances**: MD5 helps ensure the integrity of firmware and software updates.

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Applications of MD5:

Cloud Data Security:

Encryption and Hashing: MD5 is part of a hybrid approach in cloud security. For example:

- **RSA Encryption**: MD5 is used alongside RSA encryption for secure communication and file encryption/decryption.
- **MD5 Hashing**: It helps cover tables from unauthorized users and ensures data integrity.

Data Identification and Verification:

- **Unique Hashes**: Despite its flaws, MD5 consistently produces the same hash for a specific input. This property allows data identification.
- Corruption Detection: MD5 helps verify whether data has been corrupted during cloud storage or transmission

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vulnerabilities in MD5 algorithm:

Collision Attacks:

- MD5 is susceptible to **collision attacks**, where two different inputs produce the **same hash value**.
- This poses a severe security risk, especially in applications like digital signatures where uniqueness is crucial
- To overcome this collision attack **salting and peperring** is used.

Preimage Attacks

Attackers can reverse-engineer the MD5 hash to find an input that matches a given MD5 hash

Speed and Efficiency

MD5 is a **fast** hash function, which makes it vulnerable to **brute-force attacks**.

MD5's vulnerabilities have led to its decline in favor of stronger hash functions like SHA-256 or SHA-3



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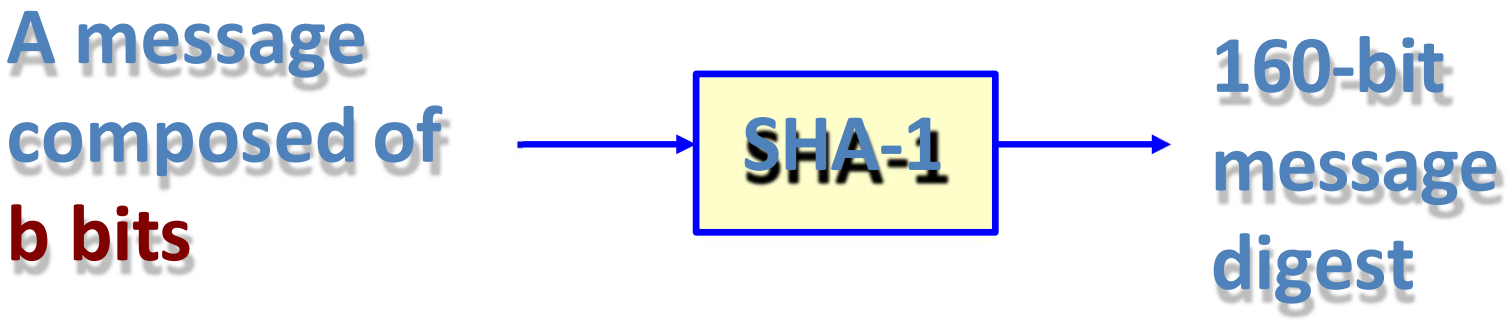
Module 2:

Lecture : SHA



SHA-1

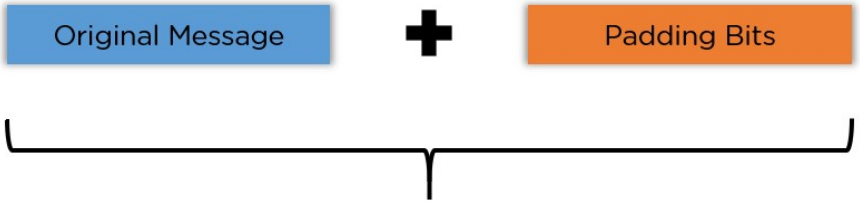
SHA works with input message of maximum length 264 bits in length and produce a message digest of length 160 bits.



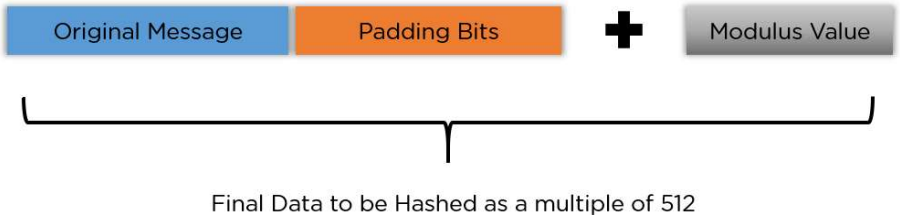
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The Working of SHA-1

Step1: Like MD5, in SHA padding of bits is done at the end of original message in such a way that total length of message is 64 bits less than the multiple of 512.



Step 2: Length of the message excluding the length of the padding is calculated and appended to the end of padding as 64-bit.



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Step 3: Divide the input into 512-bit blocks.

Step 4: Initialize 5 buffers A to E, each size is 32 bits.

A = 01 23 45 67

B = 89 ab cd ef

C = fe dc ba 98

D = 76 54 32 10

E= C3 D2 E1 F0

Copy A, B, C, D to small a, b, c, d

Step 5: Divide each 512 bits block into 16 sub-blocks.

A constant array K[t] where t is 0 to 79

Each of the sub-blocks are denoted as W[0] -> W[64].

W[0]-W[15] is same as M[0]-M[15].

W[16]-W[64] is derived as,

$W[t]=s^1(W[t-16]$ XOR W[t-14] XOR W[t-18] XOR W[t-3])



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Step 6: Mathematically, an iteration consists of

$a = (e + \text{Process P} + s^5(a) + W[t] + K[t])$,

$b = a$,

$c = s^{30}(b)$,

$d = c$,

$e = d$

s^t=Circular left-shift of 32 bit sub block by t bits.

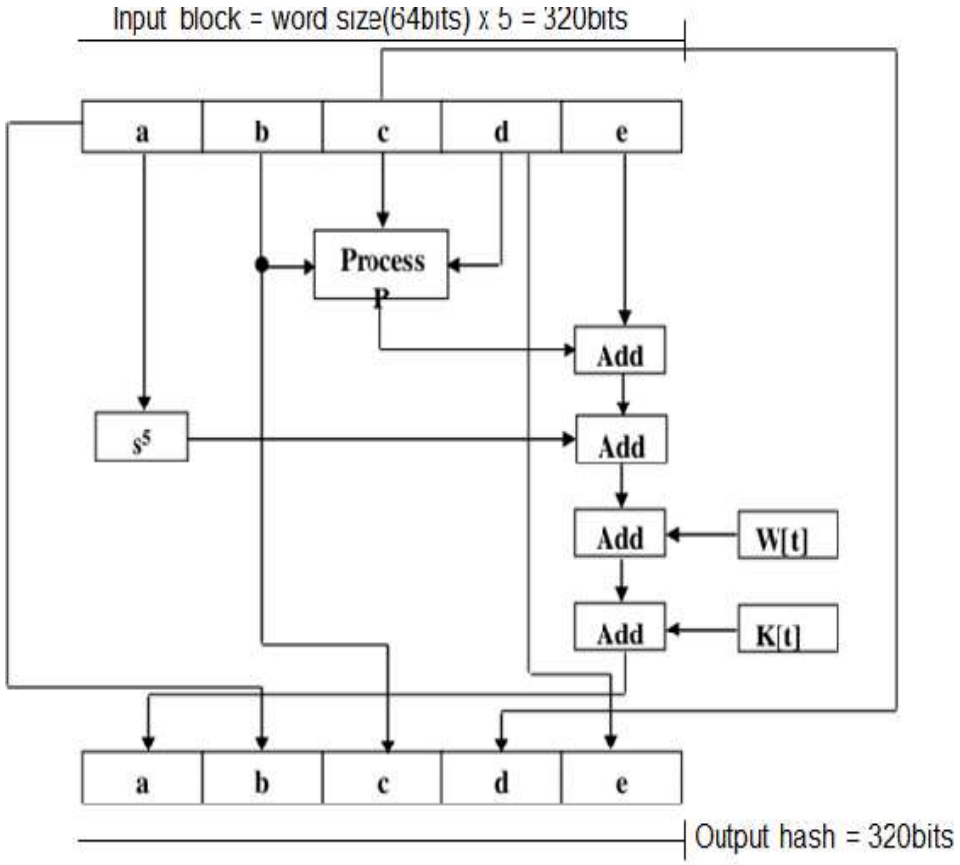


Fig 1: Modified SHA-1 Structure

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The Working of SHA-1

Round 1: (b AND c) OR ((NOT b) AND (d))

Round 2: B XOR c XOR d

Round 3: (b AND c) OR (b and D) OR (c AND d)

Round 4: B XOR c XOR d

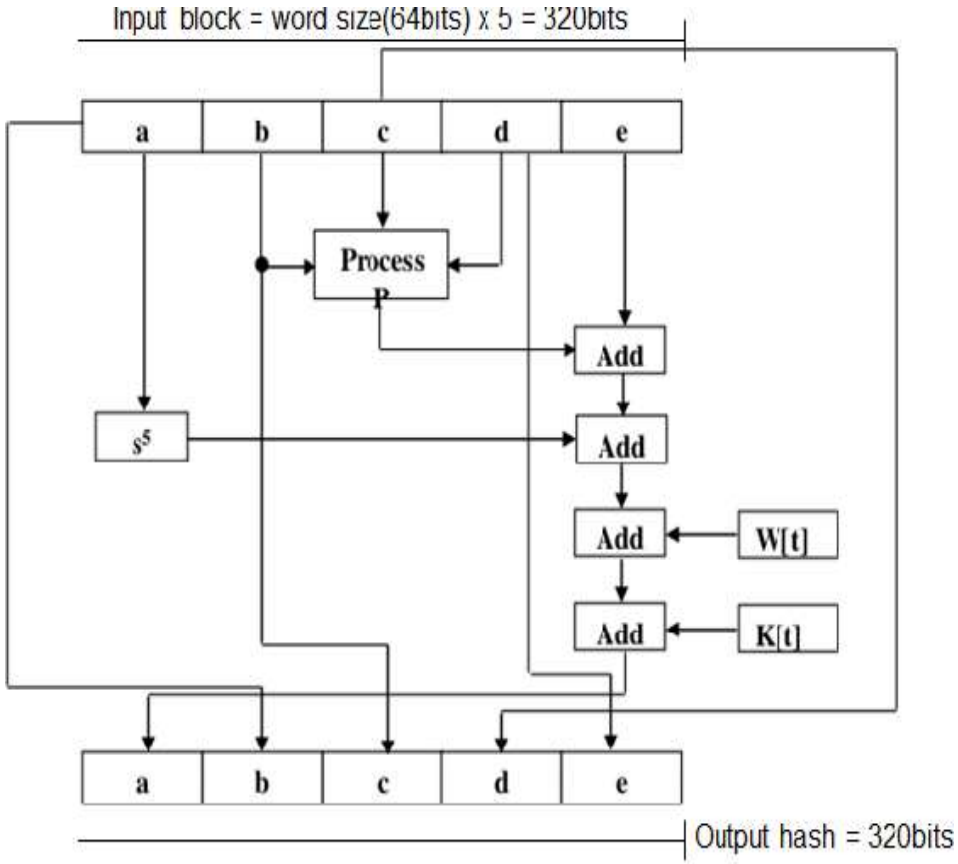


Fig 1: Modified SHA-1 Structure

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The Working of SHA-1

Table 11.3 Comparison of SHA Parameters

	SHA-1	SHA-224	SHA-256	SHA-384	SHA-512
Message Digest Size	160	224	256	384	512
Message Size	$< 2^{64}$	$< 2^{64}$	$< 2^{64}$	$< 2^{128}$	$< 2^{128}$
Block Size	512	512	512	1024	1024
Word Size	32	32	32	64	64
Number of Steps	80	64	64	80	80

Note: All sizes are measured in bits.

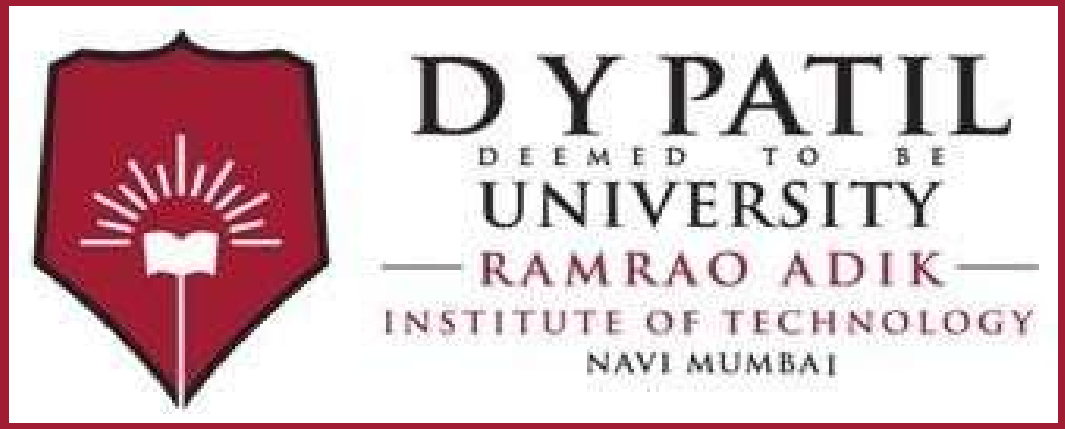
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MD5 Vs SHA

Keys For Comparison	MD5	SHA
Security	Less Secure than SHA	High Secure than MD5
Message Digest Length	128 Bits	160 Bits
Attacks required to find out original Message	2^{128} bit operations required to break	2^{160} bit operations required to break
Attacks to try and find two messages producing the same MD	2^{64} bit operations required to break	2^{80} bit operations required to break
Speed	Faster, only 64 iterations	Slower than MD5, Required 80 iterations
Successful attacks so far	Attacks reported to some extents	No such attach report yet



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Thank You